Platforms Overview

APPN Aerial SOP — Locked revision 1.0

APPN Field Platforms Overview

APPN field phenotyping platforms provide a versatile suite of UAV-based sensor technologies and associated positioning and environmental instrumentation to support crop phenotyping projects at virtually any location. From paddock-scale trials to detailed plot- and plant-level analysis, APPN platforms support **data capture, management, and analysis** across diverse research and industry use cases.

APPN M3M — Multispectral & RGB UAV Phenotyping Platform

The APPN M3M platform is a dedicated agricultural UAV system based on the **DJI Mavic series**, integrating a compact multispectral sensing suite with high-resolution RGB imagery and centimetre-accurate RTK positioning.

The M3M imaging payload includes:

- Multispectral camera with 5 MP sensors capturing **blue, green, red, red-edge, and near-infrared** bands
- Integrated **20 MP RGB camera** for visual interpretation and mapping
- Onboard **RTK GNSS** for high positional accuracy

The system is lightweight, agile, and easily deployable, with long flight endurance relative to payload weight. These characteristics make the M3M particularly well suited to:

- Rapid screening of large numbers of trials
- Repeated monitoring through the growing season
- Remote or logistically constrained field locations

See also: M3M Fieldbook.

APPN GOBI — RGB, VNIR & LiDAR UAV Phenotyping Platform

The APPN GOBI platform centres on the **GRYFN GOBI sensor package**, providing a compact yet comprehensive instrumentation suite for **high-throughput digital phenotyping** across crop trial sites and paddock-scale experiments.

GOBI integrates co-registered sensing technologies, including:

- **VNIR hyperspectral imaging** for quantitative spectral trait analysis
- **Ouster LiDAR** for canopy structure, height, and surface modelling
- **Sony RGB camera** for high-resolution visual context

The GOBI payload can be deployed on either:

- **DJI M350** — offering rapid deployment and operational flexibility, or
- **Inspired Flight IF1200** — enabling heavier payload capacity and larger survey areas

All configurations are linked to **RTK-based GNSS-INS positioning**, supporting robust geometric alignment between LiDAR, hyperspectral, and RGB data streams. This enables reliable extraction of physical and structural traits at scale.

See also: GOBI M350 Fieldbook and GOBI IF1200 Fieldbook.

APPN CALViS — Co-aligned LiDAR, VNIR & SWIR UAV Phenotyping Platform

The APPN CALViS platform provides a highly integrated sensing solution for detailed physiological and structural analysis, combining long-wave spectral coverage with 3-D canopy characterisation.

CALViS integrates:

- **Headwall co-aligned VNIR and SWIR hyperspectral sensors**
- **LiDAR instrumentation** for structural context and surface modelling
- High-accuracy GNSS-INS positioning

Hyperspectral coverage spanning approximately **350–2500 nm** enables analysis of phenotypic traits related to:

- Disease detection and progression
- Nutrient status and deficiency
- Abiotic stress indicators (e.g. water or heat stress)

The co-aligned sensor geometry ensures consistent spatial correspondence between VNIR, SWIR, and LiDAR observations, supporting advanced multi-modal analysis and trait modelling.

See also: CALViS Fieldbook.

APPN HiRes — High-Resolution RGB UAV Phenotyping Platform

The APPN HiRes platform is designed for applications requiring ultra-high- resolution visual imagery and precise spatial detail.

It combines:

- **DJI M350** four-rotor UAV
- **Phase One P3 120 MP RGB sensor**

This configuration delivers exceptionally high-quality RGB imagery suitable for:

- Detailed plot-level assessment
- Canopy structure and morphology studies
- Visual scoring, annotation, and validation

See also: HiRes Fieldbook.